

Chapter 7

Climate

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

India experiences which type of climate?

1. Temperate climate
2. Tropical Monsoon climate
3. Temperate Continental Climate
4. Mediterranean Climate

Answer

Tropical Monsoon climate

Question 2

The Deccan Plateau gets less rainfall than the Western Ghats because :

1. it lies in the way of South West Monsoon winds.
2. it lies in the rainshadow area of the Himalayas.
3. it lies in the rainshadow area of the Western Ghats.
4. it is triangular in shape.

Answer

it lies in the rainshadow area of the Western Ghats.

Question 3

What causes the winter rain along the Coromandel Coast?

1. Sea breeze
2. Retreating monsoon

3. South-west monsoon
4. Western disturbances

Answer

Retreating monsoon

Question 4

Which of the following is NOT a source of rainfall for Punjab and Haryana?

1. Western disturbances
2. Arabian Sea branch of South-west monsoon
3. Bay of Bengal branch of South-west monsoon
4. Retreating monsoon.

Answer

Retreating monsoon

Question 5

Which of the following factors does not affect the climate of a place?

1. Varied relief
2. The Monsoon Winds
3. Distance from the Sea
4. Presence of vegetation.

Answer

Presence of vegetation

Question 6

Western disturbances become active in northern India during which season?

1. Summer
2. Winter
3. Spring
4. Monsoon

Answer

Winter

Question 7

The summer monsoon winds blow in which direction?

1. North-East
2. South-West
3. North-West
4. South-East

Answer

South-West

Question 8

Which of the following is good for the cultivation of tea and jute?

1. Mango showers
2. Loo
3. Kalbaisakhi
4. None of the above.

Answer

Kalbaisakhi

Question 9

Where does the Arabian Sea branch join the Bay of Bengal branch?

1. In Punjab and Haryana
2. In Western Ghats
3. In Chota Nagpur Plateau
4. In the Sahyadris.

Answer

In Punjab and Haryana

Question 10

What causes the 'October Heat'?

1. High temperature and heavy rainfall.
2. Reduced temperature and heavy rainfall.
3. High temperature and humidity.
4. Reduced temperature and humidity.

Answer

High temperature and humidity

Question 11

Which state in India is the first to experience the onset of the monsoon and the last to see its withdrawal?

1. Tamil Nadu
2. Kerala
3. West Bengal
4. Andaman and Nicobar

Answer

Kerala

Question 12

Why do the interior parts of the Deccan remain dry during the Retreating monsoon?

1. Because they lie in the rainshadow of the Eastern Ghats.
2. Because they lie parallel to the Eastern Ghats.
3. Because they lie in the rainshadow of the Western Ghats.
4. Because they lie parallel to the Arabian Sea Branch.

Answer

Because they lie in the rainshadow of the Eastern Ghats.

Question 13

Climate of India is strongly influenced by

1. Trade winds
2. Loo
3. Easterly winds
4. Monsoon winds

Answer

Monsoon winds

Question 14

Which of the following factors cause the rise in the temperature of mainland of India in the month of March?

1. The land breeze blows towards the sea
2. The pressure belts shift.
3. Southward migration of the Sun from the equator.
4. Northward migration of the Sun from the equator.

Answer

Northward migration of the Sun from the equator.

Question 15

Which of the following areas get rainfall from the Retreating monsoon?

1. West Coast of India
2. East Coast of India
3. Western Ghats
4. Thar desert

Answer

East Coast of India

Answer the following questions

Question 1(a)

Name the type of climate prevailing over India. Mention any two factors responsible for it.

Answer

India has a tropical monsoon type of climate. Two factors responsible for this type of climate are-

1. The Himalayas
2. Varied Relief

Question 1(b)

State two important characteristic features of the Monsoon rainfall in India.

Answer

Two important characteristic features of the Monsoon rainfall in India are-

1. Reversal of wind system.
2. Uneven distribution of rainfall over the year.

Question 1(c)

Give a geographical reason for each of the following:

- (i) Mumbai is warmer than Kanpur in December.
- (ii) Punjab gets rain in winter.
- (iii) The Indo-Gangetic Plain gets some rainfall in the months of December and January.

Answer

(i) Mumbai is warmer than Kanpur in December because Mumbai is located near the sea and hence it experiences moderate climate throughout the year. Kanpur experiences continental type of climate where summers are extremely hot and winters are extremely cold.

(ii) Punjab gets rain in winter season due to some weak temperate cyclones from the Mediterranean Sea.

(iii) The Indo-gangetic plain gets some rainfall in the month of December and January because of the western disturbances, which originate in West Asia and in the regions near the Mediterranean sea. These westerly depressions bring winter rainfall over the Indo-Gangetic Plains.

Question 1(d)

- (i) What is 'Monsoon'?
- (ii) Name the place in India which receives the heaviest rainfall.

Answer

(i) Monsoon are periodic or seasonal reversing winds. They are divided into two wind systems — the Summer Monsoon that blow south-west and the Winter Monsoon that blow north-east.

(ii) Cherrapunji and Mawsynram in Meghalaya receive the heaviest rainfall.

Question 2(a)

Name two factors which affect the climate of a place.

Answer

Two factors which affect the climate of a place are-

1. Distance from sea or ocean
2. Altitude of the place

Question 2(b)

Mention two characteristics of the South-West Monsoon.

Answer

Two characteristics of the South-West Monsoon are-

1. This season lasts from June to September.
2. Winds blow from sea to land and are moisture laden winds.

Question 2(c)

Give a geographical reason for each of the following:

(i) When the Malabar coast is receiving heavy rainfall in July, the Tamil Nadu coast is comparatively dry.

(ii) The Northern Plains of India have a Continental type of climate.

(iii) Central Maharashtra receives less rainfall than Western Maharashtra.

Answer

(i) When the Malabar coast is receiving heavy rainfall in July, the Tamil Nadu coast is dry because it lies in the rain shadow region of Arabian sea branch and Bay of Bengal is parallel to the coast.

(ii) The Northern Plains of India have a continental type of climate because it is away from the moderating influence of the sea. Thus, the summers are extremely hot and winters are extremely cold.

(iii) Central Maharashtra lies in the rain shadow region whereas coastal Maharashtra is on the windward side of Western Ghats. Since places situated in the rain shadow region receive less rainfall, Central Maharashtra receives less rainfall than Western Maharashtra.

Question 2(d)

Study the climatic data given below and answer the questions that follow:

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
Temperature °C	23.1	24.8	26.5	29.3	32	32.8	33.1	32.1	30.5	29.3
Rainfall cm	15.3	10.1	0.3	0.1	1.3	4.5	6.1	10.2	10.5	20.1

- (i) Calculate the annual rainfall experienced by the station.
- (ii) Suggest a name of this station, giving a reason for your answer.
- (iii) Name the season during which rainfall is the heaviest.

Answer

- (i) The annual rainfall experienced by the station is 114.3 cm.
- (ii) The station is Tamil Nadu as the temperature remains moderate throughout the year and the rainfall is heaviest during the winter season.
- (iii) The season during which rainfall is the heaviest is retreating monsoon.

Question 3(a)

State two points of difference in climate between the northern and the southern zone of India due to latitude.

Answer

Two points of difference in climate between the northern and the southern zone of India due to latitude are —

1. The northern zone has cold winter season and hot summer season while the southern zone is warmer than the north but does not have a clear-cut winter season.
2. The northern zone doesn't have the mid day sun almost vertically overhead during any part of the year, while the southern zone has the mid day sun almost vertically overhead at least twice a year.

Question 3(b)

State briefly the impact of water bodies on the climate in coastal areas.

Answer

The water bodies have a great impact on the climate in coastal areas. Water bodies act as a major source of moisture to the summer monsoons and bring heavy rainfall to the whole area. Such areas experience moderate climate, neither too hot nor too cold.

Question 3(c)

Give a geographical reason for each of the following:

- (i) Variations in the type of climate in different regions of India.
- (ii) Most parts of India receive rainfall between June and September.
- (iii) South-West Monsoon withdraw from northern India in October.

Answer

- (i) India has a varied climate due to the presence of various relief features like mountains, deserts, presence of seas etc. Further the large latitudinal extent, altitude, presence of many mountain ranges and nearness to the oceans results in different climatic conditions in different regions.
- (ii) Most parts of India receive rainfall between June and September because the differential heating of land and sea during summer season causes the monsoon winds to drift towards the subcontinent.
- (iii) South-West Monsoon withdraw from northern India in October because the monsoon trough of low pressure over the Ganga plains becomes weaker due to the apparent southward movement of the sun. The low pressure trough is gradually replaced by high pressure.

Question 3(d)

- (i) What are Jet Streams?
- (ii) How do they influence the climate of India?

Answer

- (i) Jet streams are cold fast blowing winds that develop in the upper layers of the atmosphere.
- (ii) They influence the climate of India as the westerly jet stream prevails over the North Indian Plains during the winter months, while the easterly jet stream steers the tropical depression over India. These depressions play a significant role in the distribution of monsoon rainfall in the subcontinent. The highest rainfall occurs along the track of these depressions.

Question 4(a)

How does El-Nino affect the climate of India?

Answer

El-Nino increases the surface temperature of the sea and affects the movement of monsoon winds in the Indian Ocean and causes weak drought-like situation in the Indian sub continent.

Question 4(b)

State two characteristic features of North-East Monsoon.

Answer

Two characteristic features of North-East Monsoon are-

1. These winds blow from December to February.
2. These winds blow from land to sea.

Question 4(c)

Mention three important features of rainfall in India.

Answer

Three important features of rainfall in India are-

1. There is rainfall over three months and the rest of the year is mostly dry. Seventy percent of annual rainfall occurs in the rainy season.
2. The rains are mainly of relief type. The windward slopes of the mountains get more rainfall than the leeward side.
3. Only a small portion of the rainfall is received from sources other than the monsoon, like cyclonic rainfall and convectional rainfall.

Question 4(d)

Give a geographical reason for each of the following:

- (i) Pune receives less rainfall than Mumbai.
- (ii) Tropical cyclones bring heavy rainfall but temperate cyclones bring light rainfall.
- (iii) Kerala has a tropical climate while Punjab has a continental climate.

Answer

(i) Pune receives less rainfall than Mumbai because Mumbai lies on the windward side of the Western Ghats while Pune lies on the leeward side. Since places lying on the leeward side receive less rainfall, Pune receives less rainfall than Mumbai.

(ii) A tropical cyclone is mainly formed due to the development of a low pressure area over the sea, this results in high humidity and temperature which causes heavy rainfall. However, in a temperate cyclone wind speed is low and all sectors of the cyclone have different temperatures which is followed by continuous light rainfall for many days.

(iii) Kerala has a tropical climate while Punjab has a continental climate because Kerala is situated near water bodies while Punjab is situated in the interior of the subcontinent. Influences by the sea, Kerala experiences maritime climate, neither too hot nor too cold while Punjab experiences continental type of climate, very hot in summer and very cold in winter.

Question 5(a)

Distinguish between 'Burst of Monsoon' and 'Break of Monsoon'.

Answer

Burst of Monsoon	Break of Monsoon
The sudden violent onset of rainfall in the first week of June is termed as the burst of monsoon.	When the South-west monsoon fails to bring rainfall for weeks and there is a dry period in the rainy season, it is called break of monsoon.

Question 5(b)

Define:

- (i) Western Disturbances
- (ii) Mango Showers

Answer

(i) **Western Disturbances** — A characteristic feature of the cold weather season is the inflow of depressions from the west and the north-west. These low pressure systems are called western disturbances. They originate in West Asia and the regions near the Mediterranean Sea. They travel eastwards across Iran and Pakistan and reach India during the winter season. They bring the much needed winter rains over the plains and snowfall in the mountains.

(ii) **Mango Showers** — Mango showers are the pre-monsoon showers in the Indian states of Karnataka, Kerala, Konkan and Goa that help in the ripening of mangoes. These showers are also important for tea and coffee plants.

Question 5(c)

Give a geographical reason for each of the following:

- (i) The monsoon rain is unevenly distributed over India.

(ii) Excessive cold in north India during winter season.

(iii) Most of the India remains dry during the North-East Monsoon season.

Answer

(i) The monsoon rain is unevenly distributed over India because most parts of India receive relief rainfall, therefore windward sides always receive more rainfall than the leeward side. Some places like Aravali and Coromandel coast lie parallel to south west monsoon winds, hence, are not able to obstruct these wind and receive very little rainfall.

(ii) Excessive cold in north India during winter season because north India is far from the sea and the equator. Hence, it experiences continental type of climate — cold winters and hot summers.

(iii) Most of the India remains dry during the North-East Monsoon season because north-east trade winds blow from land to sea and hence, do not carry moisture necessary for rainfall.

Question 5(d)

(i) Name two regions which get very little rain from the Summer Monsoon.

(ii) Name any two local winds which blow in India during the summer season.

Answer

(i) Tamil Nadu and Rajasthan get very little rain from the Summer Monsoon.

(ii) Two local winds which blow in India during the summer season are Loo and Kalbaisakhi.

Question 6(a)

What is meant by the season of 'Retreating Monsoon'?

Answer

During the months of October-November, the south-west monsoon winds become weaker and start to retreat from the skies of North India. This phase of the monsoon is known as the retreating monsoon.

Question 6(b)

Name the months in which Retreating Monsoon rain is experienced.

Answer

Retreating Monsoon rain is experienced in the months of October and November.

Question 6(c)

Give a geographical reason for each of the following:

- (i) The northern plains of India do not freeze in winter.
- (ii) In spite of Aravali Range, Rajasthan receives scanty rainfall.
- (iii) Patna receives heavier rain than Delhi.

Answer

- (i) The northern plains of India do not freeze in winter because the Himalayas form a climatic barrier separating the Indian subcontinent from the rest of Asia. They prevent the cold Siberian winds from entering India.
- (ii) The state of Rajasthan receive scanty rainfall because it lies in the rain-shadow region of the Aravali Mountains. The Aravali Hills run parallel to the Arabian Sea branch, and thus do not form any barrier to intercept the winds. The winds thus passes through Rajasthan without causing any rainfall.
- (iii) Patna receives heavier rain than Delhi because Patna is on the eastern side as compared to Delhi and places situated on the eastern side receive heavier rainfall than the places situated on the western side of the country.

Question 6(d)

- (i) Name the seasons during which the North-East Trade Winds dominate.
- (ii) How do these winds affect the climate of India?

Answer

- (i) The North-East Trade Winds dominate during the winter season.
- (ii) North-east trade winds blow from land to sea, hence, most parts of the country remain dry. These winds may cause some amount of rainfall on the Coromandel coast. The nights are extremely cold while the days are pleasantly warm.

Question 7(a)

Mention the different sources of rain in Punjab and Tamil Nadu during the winter season.

Answer

During the winter season, Punjab receives rainfall from temperate cyclones that originate from the Mediterranean Sea. These cyclones are not of great intensity. Tamil Nadu receives rainfall

from the north east monsoon winds that blow over the Bay of Bengal and bring winter rainfall to the state.

Question 7(b)

What is meant by 'rainshadow area'? Give an example and state the mountains which are responsible for the rainshadow area.

Answer

A rain shadow area is an area of dry land that lies on the leeward side of a mountain. High mountains act as barriers for cold if they are high enough and lie in the path of rain – bearing winds. The leeward side of the mountains remains dry.

The western ghats in India are an example of mountains causing rain shadow area. The moisture laden winds from the Arabian Sea branch shed most of its moisture on the western slopes of the western ghats due to which by the time they reach the eastern slopes of western ghats the winds have less moisture and this side receives very less rainfall. For example, Mahabaleshwar situated on the Western side of western ghats, receives 250 cm of rainfall whereas Pune, only a few km away, receives less than 70 cm of rainfall.

Question 7(c)

State the benefits that are derived from the local winds that blow in summer in the following areas:

- (i) Kerala
- (ii) West Bengal

Answer

(i) During the summer season, the local hot winds blowing in Kerala are called Mango Showers. They bring rainfall which is very important for the ripening of the mangoes grown there.

(ii) During the summer season, the local winds blowing in West Bengal are called Kalbaisakhi. They bring rainfall which is important for jute and rice crops.

Question 7(d)

Give a geographical reason for each of the following:

- (i) Western Rajasthan receives no rain from the Arabian Sea branch of the South-West Monsoon winds.
- (ii) Mangalore is not cold even in the month of December.

(iii) Even in summer Shimla is cooler than Delhi.

Answer

(i) Western Rajasthan receives no rain from the Arabian Sea branch of the South-West Monsoon winds because the Arabian Sea branch of the South west monsoon blows parallel to the Aravali hills. Since the moisture-bearing winds do not strike the mountains, they do not cause any rainfall in Western Rajasthan.

(ii) Mangalore is situated along the Western Coast of India. Therefore, it experiences moderate impact of land breezes and sea breezes all through the year. This keeps the weather of Mangalore moderate i.e. neither too hot nor too cool. Hence, it is not cold even in the month of December.

(iii) Shimla is cooler than Delhi in summer because it is located at a higher altitude than Delhi and the temperature decreases with the height at a rate of 1°C for every 165 m of ascend.

Question 8(a)

Give two points of difference between the Arabian Sea branch and the Bay of Bengal branch of South-West Monsoon Winds.

Answer

Arabian Sea branch of South-West Monsoon Winds	Bay of Bengal branch of South-West Monsoon Winds
It enters Indian landmass after blowing over a vast open sea. During this period, it collects a lot of moisture.	It travels a shorter distance and collects a lot of moisture.
It hits against the Western Ghats as soon as it blows over the land.	It has to travel a long distance over the sea before it hits against the mountains.

Question 8(b)

Name any two states that receive rain in January-February from North-East Monsoons.

Answer

The eastern coast of Tamil Nadu and Andhra Pradesh receives much rainfall during the winter season through north-east monsoons.

Question 8(c)

Give a geographical reason for each of the following:

- (i) Western part of Rajasthan is a desert region.
- (ii) Kochi has a lower annual range of temperature than Agra.
- (iii) The Northern plains and Punjab receive winter cyclonic rain in winter.

Answer

- (i) The western part of Rajasthan is a desert because the Aravali range lies parallel to the moisture laden winds coming from the Arabian Sea. Since they do not obstruct the winds, it does not rain in western Rajasthan.
- (ii) Kochi has a coastal location while Agra is located away from the sea. Due to the influence of the moist winds from the sea, Kochi experiences a moderate climatic condition throughout the year whereas Agra has extreme temperature conditions resulting in high annual range of temperature than Kochi.
- (iii) The Northern plains and Punjab receive winter cyclonic rain in winter due to the western disturbances entering the Indian sub-continent from the North-west.

Question 8(d)

- (i) What causes the winter rain along Coromandel coast?
- (ii) Name a state which gets sufficient rainfall from the winter monsoon.

Answer

- (i) North East Trade winds cause the winter rain along Coromandel coast.
- (ii) Tamil Nadu gets sufficient rainfall from the winter monsoon.

Question 9(a)

Give two points of difference between Tropical Cyclones and Temperate Cyclones.

Answer

Tropical Cyclones	Temperate Cyclones
These are tropical depressions originating in the Bay of Bengal caused by the local variations of heat and moisture.	These low pressure systems originate in the regions near the Mediterranean Sea.
They are active in November and December.	They are active between December and

Question 9(b)

An Inter-School Football Match at Visakhapatnam from November 1 to 3 had to be cancelled because of bad weather. Give an explanation for this happening. Your explanation should provide the technical name of this season in India, as well as the pressure conditions over the Bay of Bengal.

Answer

An interschool Football Match at Vishakapatnam from November 1 to 3 had to be cancelled because of the cyclones. This is a season of retreating monsoon. During this season the low pressure conditions of the land are transferred to the centre of the Bay of Bengal that gives rise to cyclonic depressions.

Question 9(c)

Give a geographical reason for each of the following:

- (i) Kanyakumari experiences an equable climate.
- (ii) Chennai has a lower annual range of temperature than Lucknow.
- (iii) Mangalore experiences more rainfall than Mysore.

Answer

(i) Kanyakumari experiences equable climate because it is located near the coast, and there is not much difference between the temperatures during the summer and winter.

(ii) Chennai is located at lower latitude and is closer to the equator as compared to Lucknow. Besides Chennai is close to the sea so it experiences moderate climate whereas Lucknow is located away from the sea and hence it experiences continental type of climate. Therefore, Chennai has a lower annual range of temperature than Lucknow.

(iii) Mangalore lies on the windward side of Western Ghats and Mysore lies on the leeward side of the Western Ghats. Since areas located on the windward side receive more rainfall than the areas located on the leeward side, Mangalore experiences more rainfall than Mysore.

Question 9(d)

- (i) Name one region which gets rainfall from the Retreating Monsoon.
- (ii) Name two areas which get more than 200 cm of rainfall in India.

Answer

- (i) Tamil Nadu gets rainfall from the Retreating Monsoon.

(ii) Meghalaya and Arunachal Pradesh get more than 200 cm of rainfall in India.

Question 10(a)

From where does the winter showers in Western Uttar Pradesh and Punjab originate?

Answer

Winter showers in Western Uttar Pradesh and Punjab originate from the weak temperate cyclones from the Mediterranean Sea.

Question 10(b)

State the economic importance of Kalbaisakhis in West Bengal and Assam.

Answer

Kalbaisakhis are of economic importance in West Bengal and Assam because they bring rainfall in these regions. This rainfall is quite beneficial for growing jute and rice in West Bengal and tea in Assam.

Question 10(c)

Give a geographical reason for each of the following:

- (i) Chennai receives less rain than Thiruvananthapuram although it has more rainy days.
- (ii) Shillong gets less than 200 cm of rainfall in a year while Cherrapunji receives more than 1250 cm of rainfall.
- (iii) Jaipur has a higher annual range of temperature than Mumbai.

Answer

(i) Since Thiruvananthapuram lies on windward side of Western ghats therefore it receives heavy rainfall from South West monsoon while Chennai lies on the leeward side and gets rainfall from North East monsoon which is not as strong. Thiruvananthapuram also gets some rain from the retreating Monsoon.

(ii) Cherrapunji lies on the crest of the southern range of Khasi hills. It receives rain from the South West Monsoon Bay of Bengal Branch as it is in the pathway of these winds. Hence, it receives heavy rainfall. Shillong is located on the upper (leeward) side of the Garo-Khasi-Jaintia hills, which is a rain shadow area and hence, receives less than 200 cm of rainfall in a year.

(iii) Jaipur has a higher annual range of temperature than Mumbai because Jaipur is situated in the northern part of India, away from the sea. Hence, Jaipur experiences continental type of

climate, too hot in summer and too cold in winter. Mumbai, on the other hand, is situated near the sea so it has a moderate type of climate, neither too hot nor too cold.

Question 10(d)

Briefly explain the role of the Himalayas in determining the climate of India during:

- (i) the hot weather season
- (ii) the southwest monsoon season

Answer

(i) The Himalayas form a climatic barrier separating the Indian subcontinent from the rest of Asia. They prevent the cold Siberian winds from entering the Indian region and from India becoming a cold desert.

(ii) During the South West Monsoon season, the Himalayas play an important role in determining the amount of rainfall received by several places in the country. They act as a barrier to the South West Monsoon winds coming from the Indian Ocean. These winds when strike the mountains bring rainfall to the northern plains.